

Programming Assignment # 5, Clustering

Instructor: Dr. Ali Hassan

Due Date: 26 Dec 2025

Galaxy Clustering & Classification

1 Learning Objectives

Students will apply multiple clustering algorithms to real astronomical datasets using Python. The focus is on clustering and classifying galaxies using publicly available data. You will do following in this assignment:

- Apply multiple clustering algorithms to real-world astronomical datasets.
- Perform feature engineering and dimensionality reduction.
- Evaluate clustering performance using internal and external metrics.
- Develop scientific interpretations of galaxy clusters.
- Produce reproducible and well-documented Python code.

2 Dataset

You must use at least one publicly available astronomical dataset:

- Sloan Digital Sky Survey (SDSS)
- Galaxy Zoo dataset
- Kaggle Galaxy Zoo Competition Data

3 Example Cluster Structures

Below are example cluster patterns that often appear in synthetic data. You may use similar structures for sanity checks before applying algorithms to real galaxy datasets.

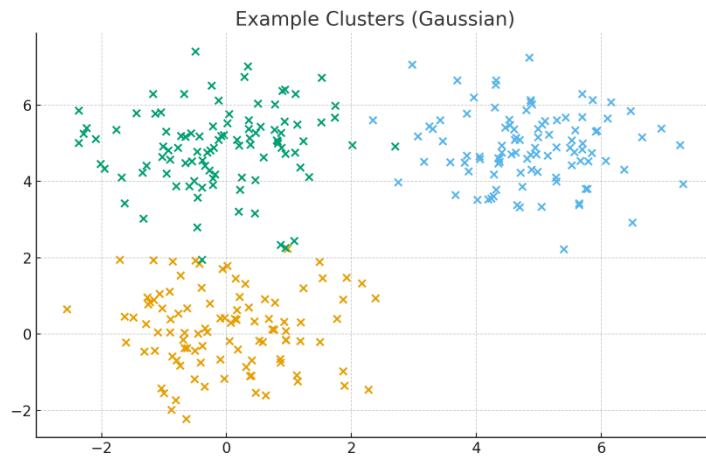


Figure 1: Example of Gaussian clusters typically used for benchmarking K-Means, GMM, or hierarchical clustering.

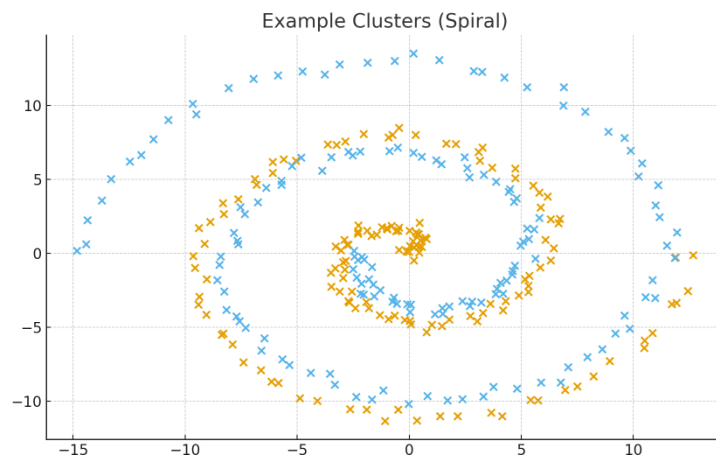


Figure 2: Example of non-linearly separable spiral clusters, useful for evaluating DBSCAN, spectral clustering, or manifold learning methods.

4 Assignment Tasks

4.1 Data Acquisition & Preprocessing

- Download or query dataset.
- Perform cleaning and exploratory data analysis.

4.2 Feature Engineering

- Construct meaningful features (e.g., photometric colors, morphology scores).

4.3 Dimensionality Reduction

- Apply PCA and at least one non-linear embedding (t-SNE, UMAP, ISOMAP, etc.)

4.4 Clustering Methods

- Apply at least four algorithms such as:
 - K-Means
 - DBSCAN
 - Gaussian Mixture Models (GMM)
 - Spectral Clustering
 - Agglomerative Clustering

4.5 Validation & Comparison

- Evaluate using silhouette score, Davies-Bouldin index, ARI, NMI etc.

4.6 Scientific Interpretation

- Interpret cluster results in an astrophysical context (e.g., morphological types, colors, redshift distributions).

4.7 Assignment Requirements

You must submit:

- A Jupyter Notebook containing all code and visualizations.
- A technical report (maximum 6 pages) summarizing the analysis.
- A requirements.txt or environment.yml file.
- Optional small data files (do not include large FITS archives).

WARNING

- Copying code or report is strictly prohibited. Violations will result in zero marks.